SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #33 – Retail, E-Commerce & Finance

Micro-Lending & Peer Credit Risk Assessor

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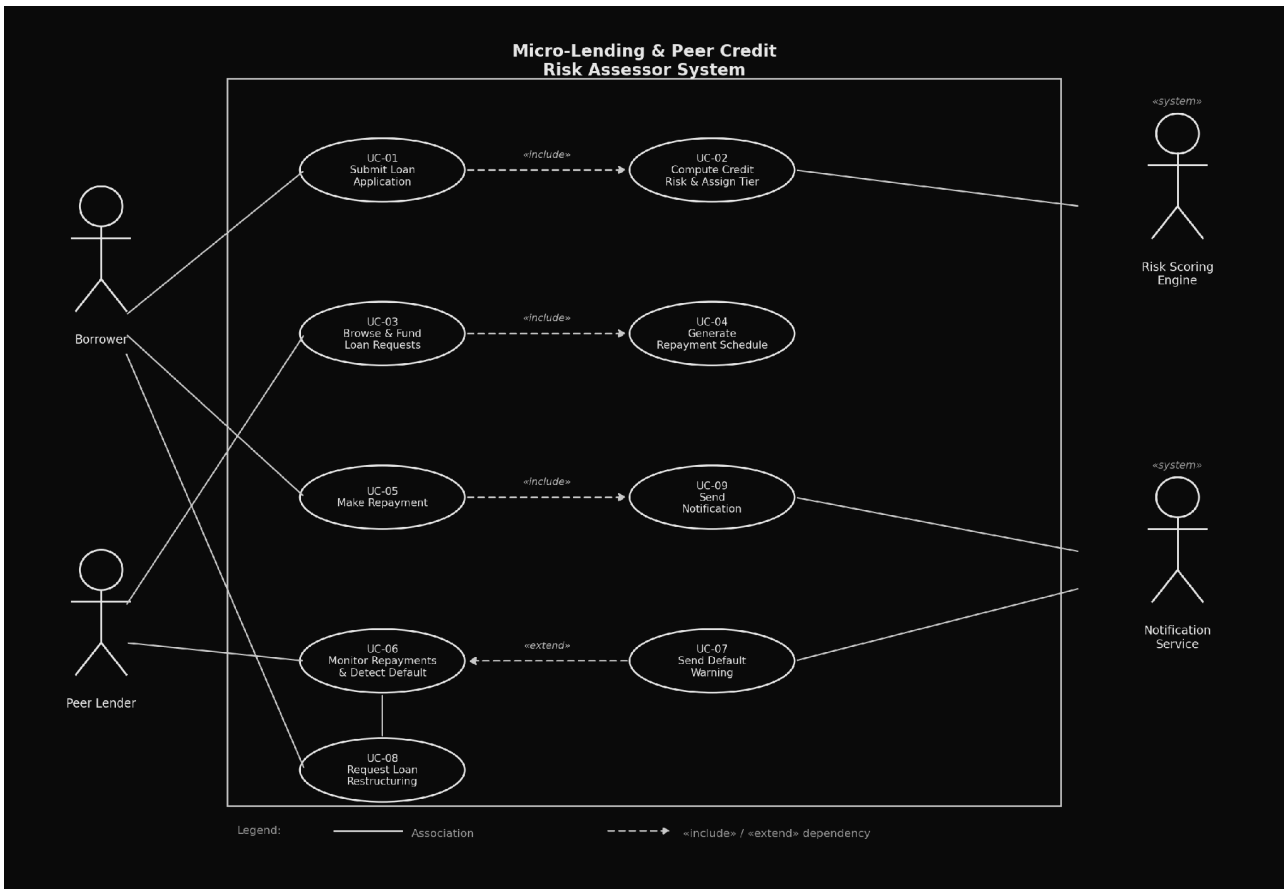
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute a Borrower's credit risk rating (Low / Medium / High) using a weighted heuristic model based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier is accurately assigned and stored against the loan application.Fail: An application with an incomplete financial profile is auto-approved or assigned a default tier.	The risk score is the foundation the entire marketplace relies on; every downstream decision (funding, pricing, default handling) depends on it being accurate and complete.
FR-002	Functional	The system shall allow a Borrower to submit a loan application specifying principal amount, purpose, and tenure, and shall reject any application missing required financial disclosure fields (income, existing debts, employment status).	High	Pass: An application submitted with all required fields is queued for risk assessment.Fail: An application missing DTI or income data is accepted into the queue.	Clean, complete borrower data is a precondition for FR-001; without it the risk engine has nothing reliable to score.
FR-003	Functional	The system shall allow a Peer Lender to browse open loan requests filtered and sorted by risk tier and offered interest rate, and to commit partial or full funding toward a request.	High	Pass: Loan status changes to "Fully Funded" once cumulative lender contributions equal the requested principal.Fail: Loan funds are disbursed before full funding is reached.	This is the core peer-to-peer matching mechanic that connects lender capital to assessed borrower risk.
FR-004	Functional	Upon a loan reaching full funding, the system shall auto-generate a structured repayment schedule (EMI amount, due dates, principal/interest breakdown) and disburse the funds to the Borrower.	Medium	Pass: A repayment schedule matching the agreed tenure and interest rate exists before/at disbursal.Fail: Funds are disbursed with no associated repayment schedule on record.	Converts committed capital into an active, trackable loan and gives both parties an auditable repayment plan from day one.
FR-005	Functional	The system shall continuously monitor repayment due dates and shall automatically trigger a default warning notification to the Borrower and all funding Peer Lenders when a scheduled EMI is missed beyond a configurable grace period.	High	Pass: A missed EMI beyond the grace period generates a notification within 1 hour and flags the loan status "At Risk".Fail: A missed payment goes unflagged after the grace period elapses.	Automated, timely default detection is the platform's core risk-management promise and lets lenders act (reminder, restructuring) before losses compound.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Non-functional	Loan repayment transactions and EMI fund distribution must execute inside database ACID transactions to prevent double-spending or partially applied fund transfers.	High	Pass: Benchmarking/load tests confirm target latency and transaction consistency under simulated peak load, with zero double-processed payments.Fail: Any test run produces a duplicated or partially applied transaction.	Financial correctness is non-negotiable on a peer-to-peer money platform; any inconsistency causes direct monetary loss and disputes between peers.
NFR-002	Non-functional	The system shall encrypt all borrower financial and personal data (income details, bank information, computed risk scores) at rest and in transit using industry-standard encryption (AES-256 at rest, TLS 1.2+ in transit).	High	Pass: A security audit / penetration test confirms no financial or PII field is stored or transmitted unencrypted.Fail: Any financial or PII field is found in plaintext at rest or in transit.	The platform holds highly sensitive peer financial data; a breach would cause severe reputational, legal and monetary harm and must be prevented by design.

2. UML Use-Case Diagram

Actors: Borrower and Peer Lender (primary/human); Risk Scoring Engine and Notification Service (supporting system actors). The diagram includes two «include» relationships (UC-01→UC-02, UC-03→UC-04, UC-05→UC-09) and one «extend» relationship (UC-07 extends UC-06).



3. Use-Case Flow Specification

Use Case ID / Name: UC-01 – Submit Loan Application

Actors: Borrower (primary), Risk Scoring Engine – system (supporting, via included use case)

Trigger: A registered Borrower chooses to request a peer-funded micro-loan.

Preconditions:

• The Borrower has an active, verified account.
• The Borrower's financial profile (income, existing debt, employment status) is available for entry.
• The lending marketplace is open for new applications.

Postconditions (Success):

The application is recorded with an assigned risk tier (Low / Medium / High) and interest rate, and is published to the marketplace as "Open for Funding".

Postconditions (Failure):

No application is published; the request status is set to "Rejected – High Risk" or "Incomplete Profile" with the specific discrepancy reason on file.

Main Success Scenario:

• 1. The Borrower logs into the platform and selects "Request a Loan."
• 2. The Borrower enters the requested principal, purpose, and tenure, and confirms/updates their financial profile (income, existing debts, employment status).
• 3. The Borrower submits the application.
• 4. The system invokes «include» UC-02 "Compute Credit Risk & Assign Tier": the Risk Scoring Engine evaluates debt-to-income ratio, repayment history, and employment stability against the heuristic model.
• 5. UC-02 returns a computed risk tier (Low / Medium / High) and a suggested interest rate to UC-01.
• 6. The system records the application, attaches the risk tier and rate, and publishes it to the marketplace with status "Open for Funding."
• 7. The system displays "Application submitted successfully" and updates the Borrower's dashboard with the assigned tier and rate.
• 8. The use case ends successfully.

Alternate Flow – A1: Application Fails Risk Assessment

A1 branches from the Main Success Scenario at step 4, when UC-02 evaluates the Borrower's profile.

• 4a1. UC-02 determines the computed risk exceeds the platform's maximum acceptable threshold (e.g., debt-to-income ratio too high, or an incomplete financial profile).
• 4a2. The system displays a rejection message listing the specific reason(s) (e.g., "Debt-to-income ratio 68% exceeds the 50% platform limit").
• 4a3. The system sets the application status to "Rejected – High Risk" / "Incomplete Profile" and does not publish it to the marketplace.
• 4a4. The use case ends without a published loan request; the Borrower may re-apply once their financial profile improves or is completed.